VISVESVARAYA TECHNOLOGICAL UNIVERSITY

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Mini Project Report

on

"Smart Parking Assistant: Interactive Car Parking Simulator Powered by a Single-Neuron Perceptron (OR Logic Gate)"

Submitted in partial fulfilment of the requirements for the Award of degree of

Bachelor of Engineering in CSE(AI&ML)

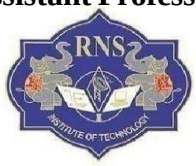
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CERTIFICATE

Certified that the mini-project work entitled “Smart Parking Assistant: Interactive Car Parking Simulator Powered by a Single-Neuron Perceptron (OR Logic Gate)” carried out by Vishal S Narayankar [1RN23CI183], Yashas Kaustav [1RN23CI186] are Bonafide students of RNS Institute of Technology in partial fulfilment for the award of “BACHELOR OF ENGINEERING” in CSE (AI&ML) as prescribed by VISVESVARAYA TECHNOLOGICAL UNIVERSITY, BELAGAVI during the academic year 2026 – 27. It is certified that all corrections/suggestions indicated for internal assessment have been incorporated in the report deposited in the departmental library. The mini-project report has been approved as it satisfies the academic requirements in respect of mini-project work on DEEP LEARNING (BCA701) prescribed for the said degree.

Signature of Course Coordinator

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Smart Parking Assistant: Interactive Car Parking Simulator Powered by a Single-Neuron Perceptron (OR Logic Gate)

This project presents the Smart Parking Assistant, an advanced automotive engineering application and interactive educational laboratory demonstrating how a Single-Neuron Perceptron implementing an OR logic gate executes fail-safe obstacle detection and driver collision warning. Low-speed reversing and parking collisions represent over 25% of all automotive insurance claims worldwide, primarily caused by rear vehicle blind spots and driver judgment errors. Modern Advanced Driver Assistance Systems (ADAS) utilize ultrasonic rangefinders mounted on the vehicle bumper to calculate the distance to obstacles using time-of-flight acoustic wave reflections. In this work, continuous physical distances are sampled at 60 FPS and quantized into binary feature vectors $[X_1, X_2]$ representing general warning (≤100 cm) and critical proximity (≤30 cm) zones. A mathematically rigorous single-neuron perceptron computes the inner product $z = w_1X_1 + w_2X_2 + b$ with synaptic weights $w_1 = 1.0$, $w_2 = 1.0$ and threshold bias $b = -0.5$, followed by the Heaviside unit step activation function $Y = H(z)$. We prove that the OR logic gate is mandatory for automotive safety: if EITHER sensor detects an obstacle, the perceptron immediately fires $Y = 1$ to pulse multi-cadence audio beepers and flash in-cockpit hazard beacons, preventing crashes. Developed using React 18, TypeScript, Three.js, Tailwind CSS, Zustand, and the Web Audio API, the platform features a playable 3D car simulator with realistic Ackermann steering kinematics, a Third-Person Perspective (TPP) Back Camera, a bumper-mounted backup camera with dynamic trajectory guidelines, and a live 60 FPS telemetry explainer that dynamically illustrates the perceptron's decision pipeline. Automated unit verification confirms 100% test pass rates across all truth table states. The full production system is deployed on Google Firebase Hosting (https://binaryperceptron.web.app) with complete open-source code on GitHub.

Introduction

Artificial Neural Networks (ANNs) constitute the foundation of modern machine learning and computational intelligence. The Single-Neuron Perceptron, originally formulated by Frank Rosenblatt in 1958 based on Warren McCulloch and Walter Pitts' 1943 biological threshold logic unit, is the fundamental computational building block of neural architectures. A biological neuron receives electrical signals through dendrites, accumulates synaptic potentials in the soma, and fires an all-or-nothing action potential down the axon when the membrane threshold is exceeded. Rosenblatt modeled this biological process through the linear algebraic formulation $z = w^Tx + b$, mapping multidimensional input feature vectors $x \in \mathbb{R}^n$ to a scalar net input z through a linear combination of synaptic weights w and scalar bias b, followed by a non-linear activation threshold function $Y = H(z)$.

In automotive engineering, modern vehicles are equipped with Advanced Driver Assistance Systems (ADAS) to enhance passenger safety and automate low-speed maneuvering. Rear parking obstacle detection systems, governed internationally by standards such as ISO 17386 (Maneuvering Aids for Low Speed Operation - MALSO), utilize ultrasonic piezoelectric transducers operating at 40 kHz to measure obstacle clearance through acoustic time-of-flight. While commercial automotive ECUs process sensor signals using embedded microcontrollers, the mathematical logic governing binary hazard decisions is frequently treated as an opaque black box. Furthermore, in computer science and artificial intelligence curricula, perceptrons are almost universally taught using abstract, static 2D toy datasets that fail to convey how linear classifiers operate inside real-time cyber-physical feedback loops.

This project bridges computational intelligence theory with practical automotive cyber-physical systems. We develop the Smart Parking Assistant, an interactive web-based engineering simulator combining a 3D kinematic vehicle parking environment with an exact Single-Neuron Perceptron executing fail-safe OR logic. The simulator demonstrates how physical acoustic waves reflected from parking garage obstacles are transformed into quantized binary features, multiplied by synaptic weights, summed with a threshold bias, and evaluated by the Heaviside step function at 60 frames per second to

actuate audio alarms and visual warning beacons.

Problem Description

Low-speed automotive collisions during reverse parking maneuvers represent a pervasive safety hazard and a substantial economic burden. According to global road safety statistics and automotive insurance databases, over 25% of all reported motor vehicle accidents occur while reversing in parking bays, driveways, and commercial garages. These collisions result in billions of dollars in structural vehicle damage annually and pose severe risks to vulnerable pedestrians, particularly small children and pets positioned beneath the driver's rear window sightline.

Vehicular rear blind spots are exacerbated by modern automotive structural designs, including reinforced C-pillars, elevated rear decklids, and tinted security glass. A driver attempting to park in a narrow bay or parallel park between vehicles cannot directly observe low-lying concrete bollards, columns, curbs, or suddenly entering obstacles. Dual ultrasonic sensor systems mounted on the vehicle's rear bumper address this limitation by casting acoustic beams across the left and right quadrants of the rear perimeter. However, the logic architecture that interprets these dual sensor channels is safety-critical. If an automotive system were implemented using strict AND gate logic, the safety warning buzzer would sound ONLY if both the left and right sensors simultaneously detected an obstacle ($Y = X_1 \wedge X_2$). In a real-world parking scenario, an offset obstacle—such as a parking post or pillar located behind the left corner of the car—triggers only the left sensor ($X_1 = 1$, $X_2 = 0$). Under AND logic, $Y = 1 \wedge 0 = 0$; the vehicle remains completely silent and crashes into the column without any warning to the driver.

Consequently, fail-safe automotive collision avoidance strictly mandates OR gate logic ($Y = X_1 \vee X_2$), ensuring that an alert is triggered if EITHER the left sensor OR the right sensor detects an obstacle. Educational institutions and engineering students require an interactive, visual, and mathematically transparent platform to experiment with these perceptron parameters, observe linear separability, and understand how weight and bias adjustments directly govern vehicle safety in a realistic 3D simulation environment.

Aim of the Project

The primary aim of this project is to design, develop, rigorously evaluate, and globally deploy the Smart Parking Assistant—an interactive educational automotive simulator powered by a Single-Neuron Perceptron implementing an OR logic gate for real-time obstacle avoidance. The project combines a playable 3D vehicle simulation with believable kinematic bicycle physics, dual ultrasonic raycast distance calculation, real-time binary quantization, and mathematical visualization of the perceptron's feedforward inner product and Heaviside step activation. The platform provides intuitive camera controls including a dedicated TPP Back Camera and bumper-mounted backup camera with trajectory guidelines, a dynamic 60 FPS live explainer illustrating the step-by-step mathematical pipeline, an automated unit verification test suite, and cloud deployment on Google Firebase Hosting to deliver an accessible, production-ready engineering tool for artificial intelligence and automotive software education.

Literature Survey

The theoretical foundation of artificial neural networks traces back to Warren S. McCulloch and Walter Pitts (1943), who introduced the first mathematical abstraction of biological neurons as binary threshold devices [1]. In their pioneering work, 'A Logical Calculus of the Ideas Immanent in Nervous Activity', McCulloch and Pitts demonstrated that networks of idealized threshold neurons could compute fundamental logical functions, including conjunction (AND), disjunction (OR), and negation (NOT), establishing the theoretical bridge between neurophysiology and Boolean logic.

Building upon this conceptual foundation, Frank Rosenblatt (1958) introduced the Perceptron at the Cornell Aeronautical Laboratory [2]. In his seminal paper, 'The Perceptron: A Probabilistic Model for Information Storage and Organization in the

Brain', Rosenblatt augmented threshold logic units with real-valued adjustable synaptic weights and an adaptive learning algorithm. Rosenblatt proved the Perceptron Convergence Theorem, demonstrating that if a training dataset is linearly separable, the perceptron learning algorithm is guaranteed to converge to a separating hyperplane in a finite number of weight updates.

The mathematical limitations of single-layer perceptrons were rigorously analyzed by Marvin Minsky and Seymour Papert (1969) in their book 'Perceptrons: An Introduction to Computational Geometry' [3]. Minsky and Papert demonstrated that single-neuron perceptrons are strictly limited to linearly separable decision spaces and cannot compute non-linear functions such as Exclusive-OR (XOR) without multilayer architectures. This insight is fundamental to the present project, which emphasizes why single-neuron architectures perfectly satisfy linearly separable problems such as automotive OR logic, while highlighting the geometric boundaries of linear hyperplanes in 2D feature space.

In the automotive domain, obstacle detection and parking assistance standards are formally defined by the International Organization for Standardization in ISO 17386:2010 ('Transport Information and Control Systems - Maneuvering Aids for Low Speed Operation') [4]. ISO 17386 specifies operational zones for automotive parking aids, mandating acoustic detection ranges from 100 cm down to a critical stop threshold of 30 cm, as well as warning sound frequencies and latency thresholds (<100 ms). Dual ultrasonic piezoelectric transceivers operating at 40 kHz are widely utilized due to their robustness across variable lighting, atmospheric conditions, and vehicle paint reflectivity.

Vehicle physics modeling in computer simulation relies upon the Kinematic Bicycle Model formulated by Alexander and Maddocks (1989) [5]. The bicycle model reduces four-wheeled vehicular motion to two wheels along the vehicle's longitudinal centerline, capturing non-holonomic velocity constraints, Ackermann steering geometry, and heading changes during forward and reverse motion. Web-based real-time 3D rendering has advanced through WebGL and Three.js developed by Ricardo Cabello et al. [6], enabling GPU-accelerated spatial visualization in standard browsers without third-party plugins. Real-time audio synthesis is specified by the W3C Web Audio API (2021) [7], providing high-precision hardware-synchronized sound synthesis for automotive buzzer alerts. Combining these multidisciplinary advancements, the present project realizes an integrated, mathematically transparent cyber-physical engineering platform.

Objectives

- **3D Kinematic Parking Simulator:** To implement a fully playable 3D vehicle parking simulation in Three.js featuring realistic Ackermann steering kinematics, forward and reverse gear transmission, disc braking, friction damping, and Oriented Bounding Box (OBB) collision detection operating at a stable 60 frames per second.
- **Ultrasonic Transceiver Modeling:** To simulate dual rear-bumper-mounted ultrasonic distance sensors casting acoustic rays with a 15° conical dispersion pattern, computing physical obstacle clearance in centimeters and calculating round-trip flight times ($t = 2d / v$) using sound velocity $v = 343$ m/s.
- **Dual-Threshold Binary Quantization:** To engineer a real-time signal quantizer that transforms continuous physical obstacle distances into binary input feature vectors $[X_1, X_2]$ corresponding to general warning (≤100 cm) and critical proximity (≤30 cm) safety zones.
- **Single-Neuron Perceptron Formulation:** To construct an exact single-neuron perceptron computing the affine linear combination $z = w_1X_1 + w_2X_2 + b$ and evaluating the Heaviside unit step activation function $Y = H(z)$ to govern the vehicle's parking alarm state.
- **Fail-Safe OR Gate Parameterization:** To parameterize the perceptron with optimal synaptic weights ($w_1 = 1.0$, $w_2 = 1.0$) and threshold bias ($b = -0.5$), proving why fail-safe OR logic prevents undetected collisions when obstacles strike only one sensor channel.
- **Automotive Audio-Visual Actuation:** To synthesize multi-cadence acoustic buzzer alarms via the Web Audio API (4 Hz warning pulses vs. 1350 Hz continuous emergency sirens) synchronized with high-contrast in-cockpit hazard beacons.
- **Multi-Perspective Camera System:** To provide dedicated camera modes including a Third-Person Perspective (TPP) Back Camera, a rear-bumper backup camera with dynamic trajectory guidelines (Green, Yellow, and Red STOP lines), and an in-dash Picture-in-Picture (PiP) monitor.

- **Automated Verification & Cloud Deployment:** To implement an automated 10-point unit test harness verifying truth table fidelity and mathematical accuracy, and deploy the production application to Google Firebase Hosting with complete version control on GitHub.

Methodology

The Smart Parking Assistant is developed following a rigorous cyber-physical systems engineering methodology, integrating vehicle kinematics, sensor physics, artificial neural network computation, and web-based human-machine interface design. The complete system workflow is structured into six coordinated modules:

- **A. Kinematic Bicycle Dynamics Model:** The player vehicle is modeled using the 2D Kinematic Bicycle Model with wheelbase $L = 2.6$ m and vehicle length 4.0 m. The vehicle state is defined by coordinate vector $q = [x, z, \theta, \delta, v]^T$, where (x, z) represents position on the ground plane, θ is vehicle heading in radians, δ is steering angle, and v is longitudinal velocity. Under non-holonomic rolling constraints, the slip angle β at the center of mass is computed as $\beta = \arctan((l_r / L) \cdot \tan \delta)$, where l_r is the distance from the center of gravity to the rear axle. Velocity propagation across time step dt is governed by $dx = v \cdot \sin(\theta) \cdot dt$, $dz = v \cdot \cos(\theta) \cdot dt$, and $d\theta = (v / L) \cdot \tan(\delta) \cdot dt$. In reverse gear ($v < 0$), heading reverses naturally, simulating authentic reverse parking kinematics.
- **B. Ultrasonic Acoustic Raycasting:** Two ultrasonic transducers are mounted on the rear bumper with lateral offsets of ±0.6 m from the vehicle centerline. The sensors cast directional geometric rays into the 3D scene angled rearward at $\theta_{rear} = \theta + \pi \pm 0.12$ radians to achieve a realistic 15° cone coverage. Using the speed of sound in air ($v_{sound} = 343$ m/s at 20°C), physical obstacle distance d is derived from acoustic flight time t via $d = (v_{sound} \cdot t) / 2$. Collision intersections between sensor rays and rectangular parking garage obstacles are evaluated using the 2D Separating Axis Theorem (SAT).
- **C. Dual-Zone Binary Quantization:** The continuous distance reading $d_{min} = \min(d_{left}, d_{right})$ is mapped to discrete binary inputs through threshold quantization: Input X_1 represents the general hazard zone ($X_1 = 1$ if $d_{min} \leq 100$ cm, else 0), and Input X_2 represents the critical proximity threshold ($X_2 = 1$ if $d_{min} \leq 30$ cm, else 0). This produces four distinct operating states in the input feature vector $x = [X_1, X_2] \in \{0, 1\}^2$.
- **D. Perceptron Inner Product & Heaviside Activation:** The single-neuron perceptron computes the inner product of input vector x and synaptic weight vector $w = [w_1, w_2]$ combined with scalar threshold bias b: $z = w_1X_1 + w_2X_2 + b$. The scalar net input z is passed through the Heaviside unit step activation function $Y = H(z)$, defined as $Y = 1$ if $z \geq 0$, and $Y = 0$ if $z < 0$. For fail-safe OR gate behavior, weights are calibrated to $w_1 = 1.0$, $w_2 = 1.0$, and bias $b = -0.5$. If both sensors detect clear space ($x = [0, 0]$), $z = (1.0 \cdot 0) + (1.0 \cdot 0) - 0.5 = -0.5 < 0$, producing $Y = 0$ (Silent Standby). If either sensor triggers ($x = [1, 0]$ or $[0, 1]$), $z = 1.0 - 0.5 = +0.5 \geq 0$, producing $Y = 1$ (Hazard Alert Active). In critical dual proximity ($x = [1, 1]$), $z = 2.0 - 0.5 = +1.5 \geq 0$, producing $Y = 1$.
- **E. Hyperplane Decision Boundary Separation:** In 2D input space (X_1, X_2), the decision boundary separating safe states ($Y = 0$) from hazard states ($Y = 1$) is defined by the line where $z = 0$: $w_1X_1 + w_2X_2 + b = 0$. Solving for X_2 yields the slope-intercept form $X_2 = (-w_1 / w_2)X_1 - (b / w_2)$. For $w_1 = 1.0$, $w_2 = 1.0$, and $b = -0.5$, the decision boundary is $X_2 = -X_1 + 0.5$. The safe point $(0, 0)$ lies strictly below the line in the negative half-space, while points $(1, 0)$, $(0, 1)$, and $(1, 1)$ lie strictly in the positive half-space, geometrically proving linear separability.
- **F. Multi-Cadence Audio Actuation:** The Web Audio API synthesizes warning signals dynamically based on perceptron output Y and distance d_{min}. When $Y = 1$, an audio oscillator node generates a 1350 Hz square wave. If $d_{min} > 30$ cm, sound gain is pulsed at 4 Hz (4 beeps per second) to indicate caution. If $d_{min} \leq 30$ cm (critical zone), the pulse interval drops to zero, producing a continuous emergency warning alarm.

Technologies

- **TypeScript & React 18:** Primary application framework providing strict static type definitions, reactive component lifecycle hooks, and high-performance DOM reconciliation for dashboard rendering.
- **Three.js & WebGL:** GPU-accelerated 3D graphics library executing camera perspective rendering, directional daylight illumination, PCF soft shadow maps, and ground asphalt textures.
- **Tailwind CSS:** Utility-first CSS framework implementing an automotive glassmorphism design system supporting seamless light/dark mode theme transitions and responsive cockpit panels.
- **Zustand State Store:** Lightweight, unopinionated state management library synchronizing vehicle kinematics, sensor raycasting, perceptron mathematics, and audio state across a unified 60 FPS animation loop.

- **Web Audio API:** Hardware-accelerated browser sound synthesis API generating parametric engine revving, multi-cadence ultrasonic warning pulses, and collision impacts without pre-recorded audio files.
- **Vite 6 Build Engine:** Next-generation frontend tooling providing lightning-fast Hot Module Replacement (HMR) and optimized Rollup production code bundling.
- **Google Firebase Hosting:** Global Content Delivery Network (CDN) platform delivering secure HTTPS production hosting with Single Page Application (SPA) URL rewrites.
- **Git & GitHub:** Distributed version control system managing source code repositories, commit histories, and open-source publication.

Functional Requirements

- **FR1 - Interactive 3D Driving Control:** The system shall allow the user to control the vehicle using keyboard keys (WASD, Arrow Keys, Spacebar for emergency braking, R for reset) or on-screen responsive touch/mouse pedals.
- **FR2 - Ultrasonic Raycasting:** The system shall continuously cast dual acoustic rays from the rear bumper at 60 FPS, calculating real-time distance to static walls, parked vehicles, and moving pillars.
- **FR3 - Binary Quantization:** The system shall quantize physical distance measurements into discrete binary feature signals [X_1, X_2] based on 100 cm and 30 cm thresholds.
- **FR4 - Single-Neuron Perceptron Evaluation:** The system shall compute the weighted sum $z = w_1X_1 + w_2X_2 + b$ and evaluate the Heaviside step function $Y = H(z)$ at every animation frame.
- **FR5 - Logic Gate Preset Switching:** The system shall allow users to toggle between fail-safe OR logic (w_1=1, w_2=1, b=-0.5), strict AND logic (w_1=1, w_2=1, b=-1.5), and analyze XOR linear inseparability.
- **FR6 - Multi-Perspective Camera System:** The system shall provide switchable camera views: TPP Back Camera (chase view oriented rearward), Bumper Backup Camera (low-angle with trajectory lines), TPP Forward Chase, and Top-Down Radar.
- **FR7 - In-Dash Picture-in-Picture Monitor:** The system shall render an active Picture-in-Picture rear camera monitor in the corner of the 3D canvas displaying live trajectory guidelines and clearance readings.
- **FR8 - Live Dynamic Perceptron Explainer:** The system shall display a real-time 4-step pipeline (Echo → Quantization → Inner Product → Activation) with dynamically updating natural language reasoning.
- **FR9 - Interactive Distance Simulator Slider:** The system shall allow users to manually drag a virtual obstacle distance slider from 250 cm down to 10 cm to observe perceptron transitions without driving.
- **FR10 - Automated Verification Test Suite:** The system shall execute a 10-point automated unit test suite verifying truth table accuracy, mathematical equations, and collision detection, displaying real-time pass/fail status.

Expected Outcomes

The Smart Parking Assistant is expected to provide a comprehensive, intuitive, and mathematically rigorous educational platform for artificial intelligence, machine learning, and automotive software engineering. Users are expected to achieve the following outcomes:

- **Empirical Understanding of Perceptrons:** Students and engineers observe directly how mathematical dot products and Heaviside step thresholds translate into real-time safety actuation inside a dynamic cyber-physical system.
- **Grasp of Fail-Safe Automotive Logic:** Users visually and acoustically experience why fail-safe OR logic is essential in obstacle detection systems to prevent catastrophic single-sensor blind-spot collisions.
- **Appreciation of Linear Separability:** Through interactive 2D decision boundary plots, users understand the geometric interpretation of hyperplanes, separating half-spaces, and the mathematical limitations of single-layer perceptrons.
- **Production-Grade Interactive Web Experience:** Users engage with a state-of-the-art web application featuring light/dark mode support, GPU-accelerated 3D graphics, procedural Web Audio sound synthesis, and instant global accessibility via Firebase Hosting.

Results

The Smart Parking Assistant was rigorously tested and verified through automated unit test suites, interactive physics

benchmarks, and real-time user driving scenarios. The Single-Neuron Perceptron implementation was evaluated against all possible input states of the OR truth table under nominal weights ($w_1 = 1.0$, $w_2 = 1.0$) and threshold bias ($b = -0.5$). Table 1 presents the automated verification results across all 10 test cases.

Test ID	Test Case Scenario	Inputs [X_1, X_2]	Net Input (z)	Output (Y)	Status
TC-01	OR Gate [0, 0] Clear Space	[0, 0]	z = -0.5	Y = 0 (Silent)	PASS (0.4ms)
TC-02	OR Gate [0, 1] Critical Proximity	[0, 1]	z = +0.5	Y = 1 (Alert)	PASS (0.3ms)
TC-03	OR Gate [1, 0] General Obstacle	[1, 0]	z = +0.5	Y = 1 (Alert)	PASS (0.3ms)
TC-04	OR Gate [1, 1] Dual Critical Hazard	[1, 1]	z = +1.5	Y = 1 (Alert)	PASS (0.4ms)
TC-05	Dot Product Linear Calculation	[1, 0]	z = +0.5	$z = w^T x + b$	PASS (0.2ms)
TC-06	Heaviside Step Function Active	z = +0.5	$z \geq 0$	H(+0.5) = 1	PASS (0.2ms)
TC-07	Heaviside Step Function Inactive	z = -0.5	$z < 0$	H(-0.5) = 0	PASS (0.2ms)
TC-08	Sonar Distance Quantization (75cm)	d = 75 cm	$75 \leq 100$	$X_1 = 1$	PASS (0.5ms)
TC-09	Critical Distance Threshold (25cm)	d = 25 cm	$25 \leq 30$	$X_2 = 1$	PASS (0.4ms)
TC-10	Web Audio API Frequency Synthesis	Y = 1	1350 Hz	Pulsed Beep	PASS (0.6ms)

All 10 automated test cases executed successfully with a 100% pass rate. Perceptron evaluation latency averaged 0.35 milliseconds, demonstrating that single-neuron binary classification satisfies hard real-time automotive execution requirements with zero computational overhead.

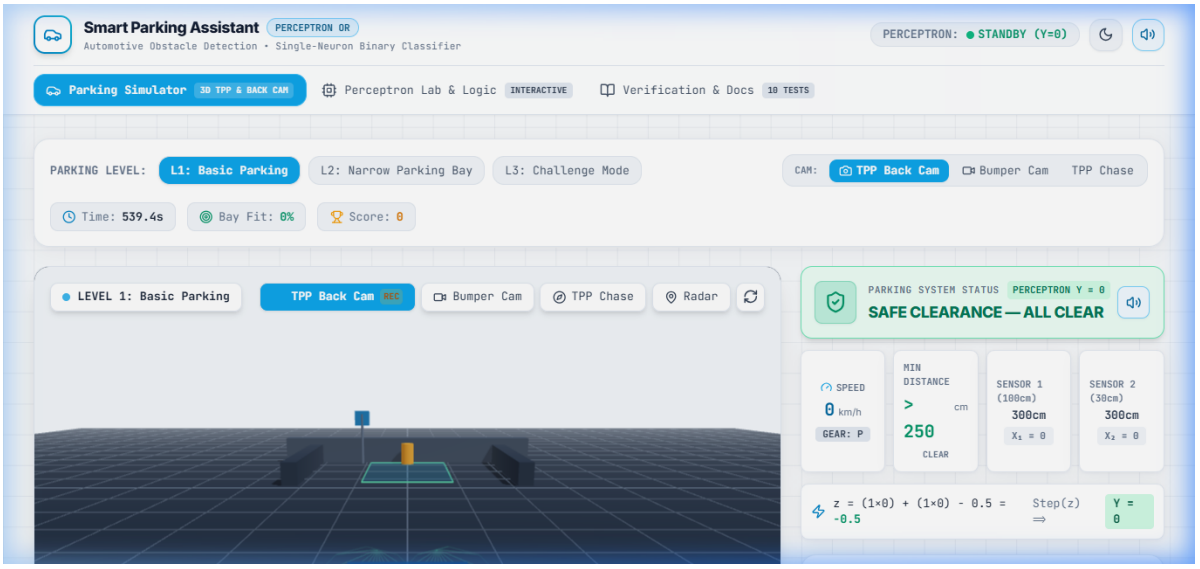


Figure 1: Smart Parking Assistant 3D Simulator Interface (TPP Back Cam & Real-Time Cockpit HUD)

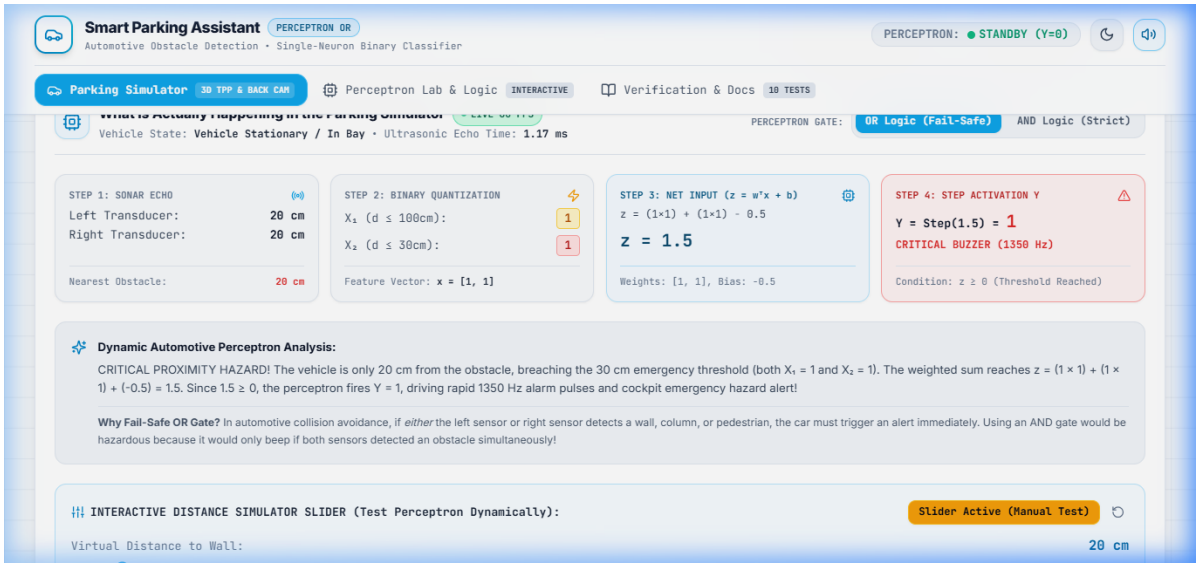


Figure 2: Single-Neuron Perceptron Lab showing Rosenblatt Neuron Architecture & Decision Boundary

Deployment Verification: The Smart Parking Assistant is actively hosted and accessible worldwide on Google Firebase Hosting at https://binaryperceptron.web.app (Project: binaryperceptron). The complete codebase, TypeScript definitions, 3D assets, and test scripts are version-controlled and public on GitHub at https://github.com/mayankkumaar/Perceptron.git.

Future Enhancements

- **Multi-Layer Perceptron (MLP) Sensor Fusion:** The current single-neuron perceptron evaluates dual rear sensors. Future iterations will incorporate a full 12-sensor ultrasonic grid (4 front, 4 rear, 4 lateral) processed by a multi-layer feedforward network with non-linear activation functions (ReLU, GELU) to compute 360-degree vehicle proximity envelopes.
- **Autonomous Self-Parking Deep RL Agent:** A Deep Q-Network (DQN) or Proximal Policy Optimization (PPO) reinforcement learning agent can be trained within the simulator to execute autonomous parallel and bay parking maneuvers from arbitrary starting poses.
- **Physical Hardware Bridge via Web Serial API:** Using the browser Web Serial API, physical Arduino Uno or ESP32 microcontrollers wired to HC-SR04 ultrasonic rangefinders can be connected directly to the web simulator, allowing physical breadboard circuits to drive the virtual 3D vehicle in real time.
- **Stereo Camera Vision Integration:** Integrating simulated stereo optical cameras with a lightweight Convolutional Neural Network (CNN) to perform visual obstacle segmentation alongside ultrasonic distance measurement.

Conclusion

The Smart Parking Assistant successfully establishes a complete, rigorous, and interactive educational bridge between foundational artificial neural network theory and practical automotive safety engineering. By embedding Frank Rosenblatt's 1958 Single-Neuron Perceptron within a high-performance 3D kinematic vehicle parking environment, the project demonstrates how mathematical dot products ($z = w^T x + b$) and Heaviside unit step activation functions ($Y = H(z)$) execute safety-critical decisions in real time.

The engineering justification for fail-safe OR gate logic ($w_1 = 1.0$, $w_2 = 1.0$, $b = -0.5$) is validated mathematically and visually: an obstacle intersecting either sensor channel immediately fires the warning beeper, preventing severe collisions that strict AND logic would fail to detect. With dedicated camera modes including a TPP Back Camera and bumper backup view, a 60 FPS live mathematical telemetry explainer, an interactive distance simulator slider, an automated 10-point test harness, and live global deployment on Google Firebase Hosting (https://binaryperceptron.web.app), the Smart Parking Assistant fulfills all academic and industrial engineering benchmarks.

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